Introduction to Urban Data Science

CRP 4680/5680 Spring 2025

Week 13 Machine Learning (I)
--Supervised Learning Algorithm

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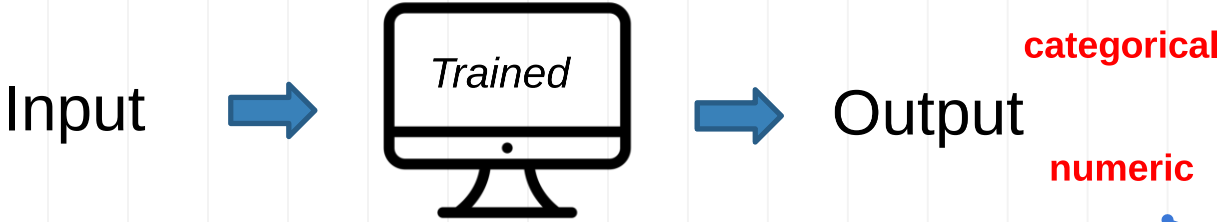
Python and Machine Learning

Supervised Learning

- **Two tasks :**
  - Classification: output is **categorical**
    - Spam email or not
    - With a table or not
  - Regression: output is **numeric**
    - Housing price

Logistic Regression is here!!!
Decision Trees; Random Forest; Support Vector Machines (SVM); k-Nearest Neighbors (k-NN)

Linear Regression is here!!!
Polynomial Regression
Ridge Regression / Lasso Regression
XGBoost



Algorithm Example: Linear Regression

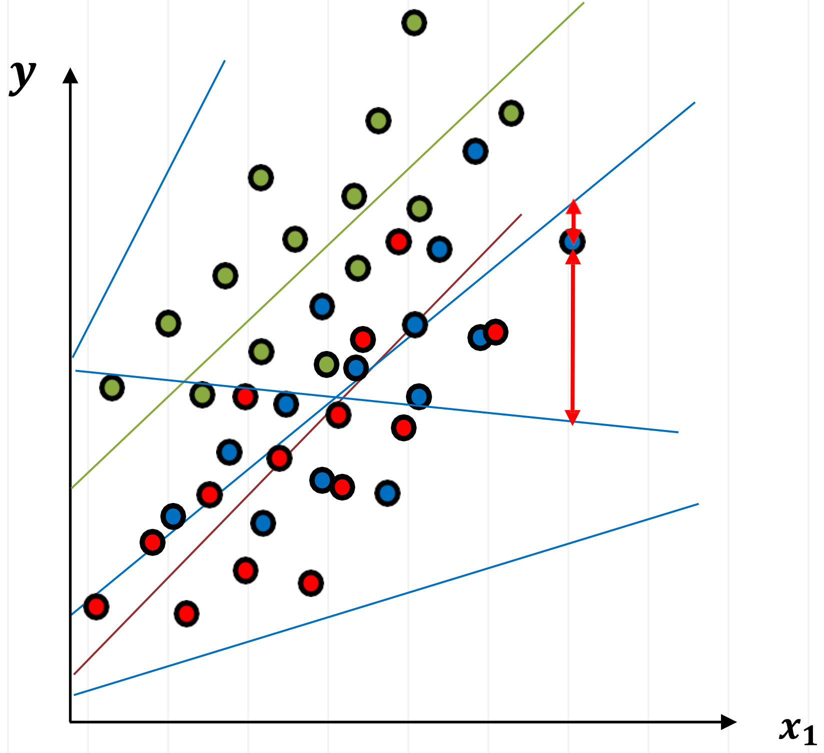
- **Supervised learning**
  - Regression Problem

y	x_1
20000	800
60000	2000
14000	632
54000	1800
20000	750
20000	400
…	…

$$y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i$$

OLS: Ordinary Least Squares

minimize sum of squared errors

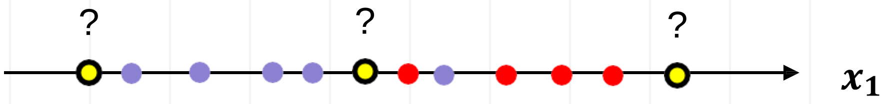


Algorithm Example: Classification

- y = 0
- y = 1

- **Supervised learning**
  - Classification Problem
    - ❖ Binary Choice

y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…



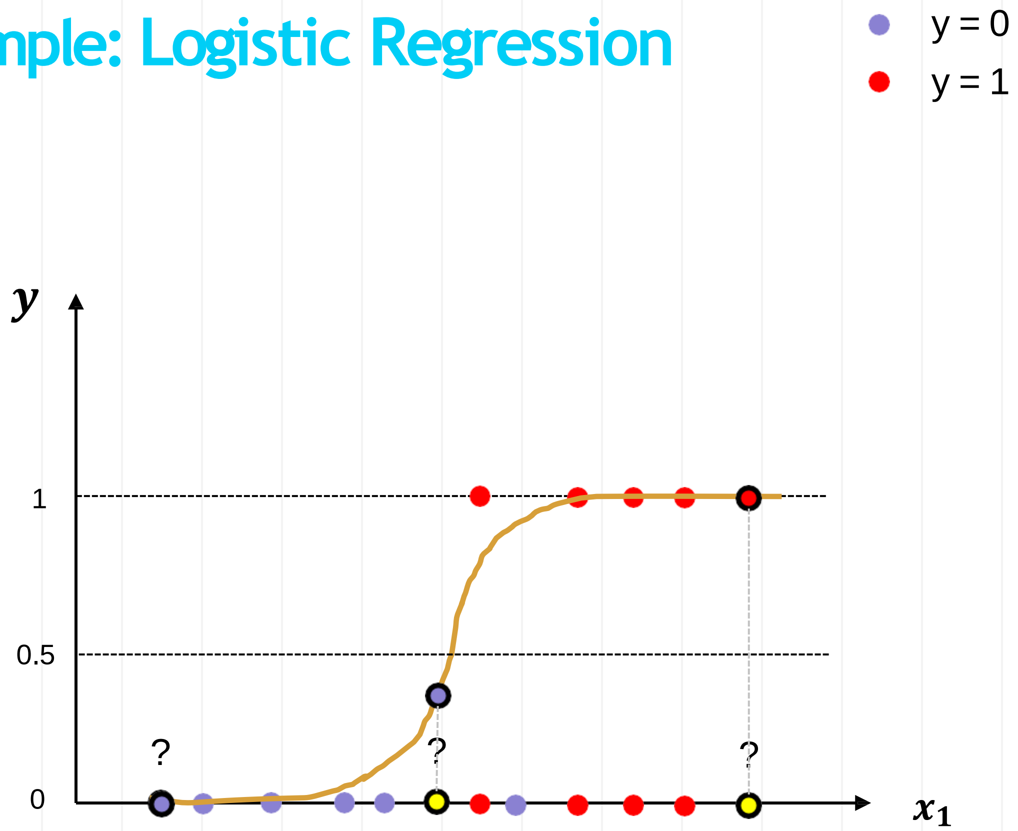
Algorithm Example: Logistic Regression

- **Supervised learning**
  - Classification Problem
    - Binary Choice

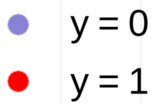
y	x_1
0	800
1	2000
0	632
1	1800
1	750
0	400
…	…

$$P(y_i = 1|x_{1i}) = \frac{\exp(\beta_0 + \beta_1 x_{1i})}{1 + \exp(\beta_0 + \beta_1 x_{1i})}$$

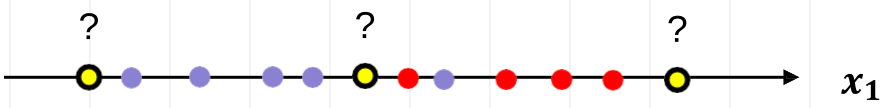
MLE: Maximum Likelihood Estimation



Algorithm Example: K-Nearest Neighbors



- **Supervised learning**
  - Classification Problem
    - Binary Choice



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…	…

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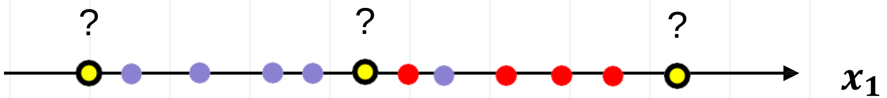
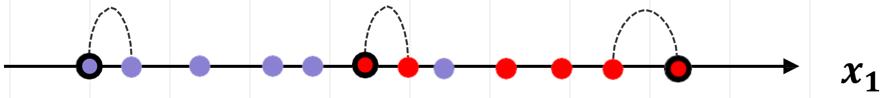


1 Nearest Neighbor



Algorithm Example: K-Nearest Neighbors

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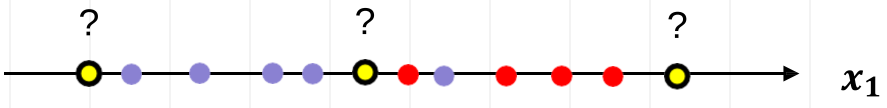
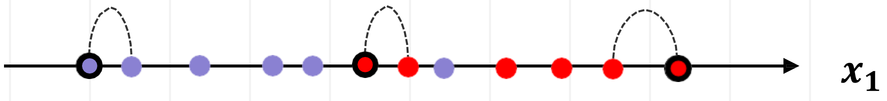
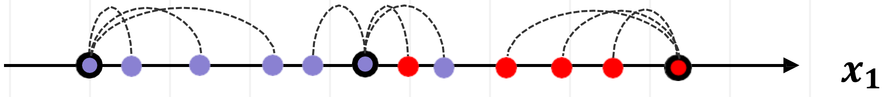


1 Nearest Neighbor



3 Nearest Neighbor



k Nearest Neighbor…

Algorithm Example: K-Nearest Neighbors

- **Supervised learning**
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    - ❖ Binary Choice

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1	750
0	400
…	…

KNN is often referred to as a "lazy learner“

- it simply stores the training data and makes predictions by calculating distances to the known data points

The number of nearest neighbors (k) to consider when making a prediction

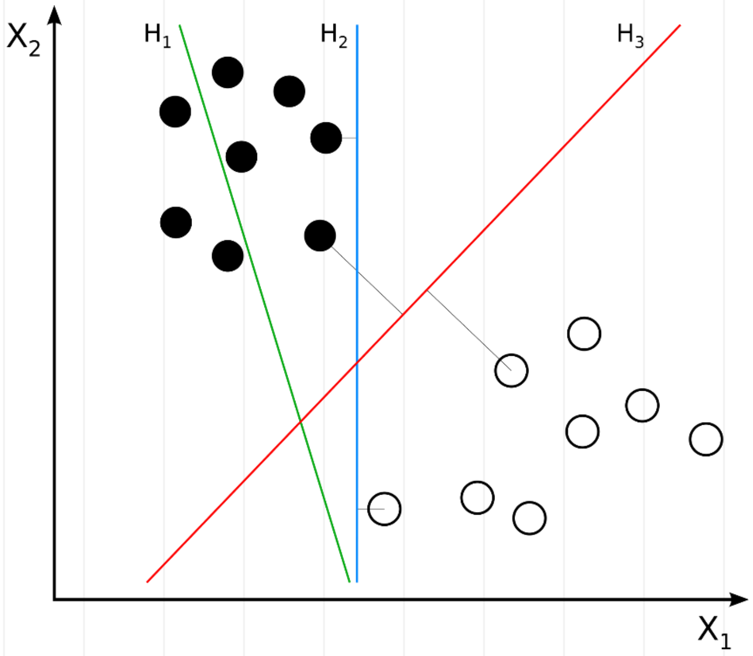
- **Small k:** Can lead to a model that is sensitive to noise in the training data (overfitting).
- **Large k:** Can lead to a model that is too generalized (underfitting).

Support Vector Machine

- **Objective**: To find a line or plane (in higher dimensions) that maximizes the distance between the line/plane and the nearest training data points of any class

- Key Concept:
  - The maximum-margin **hyperplane**.
  - **Margin:** the distance between the hyperplane and the closest data points from either class.
  - These closest points are called **support vectors**.

- margin is maximized equally for both classes.



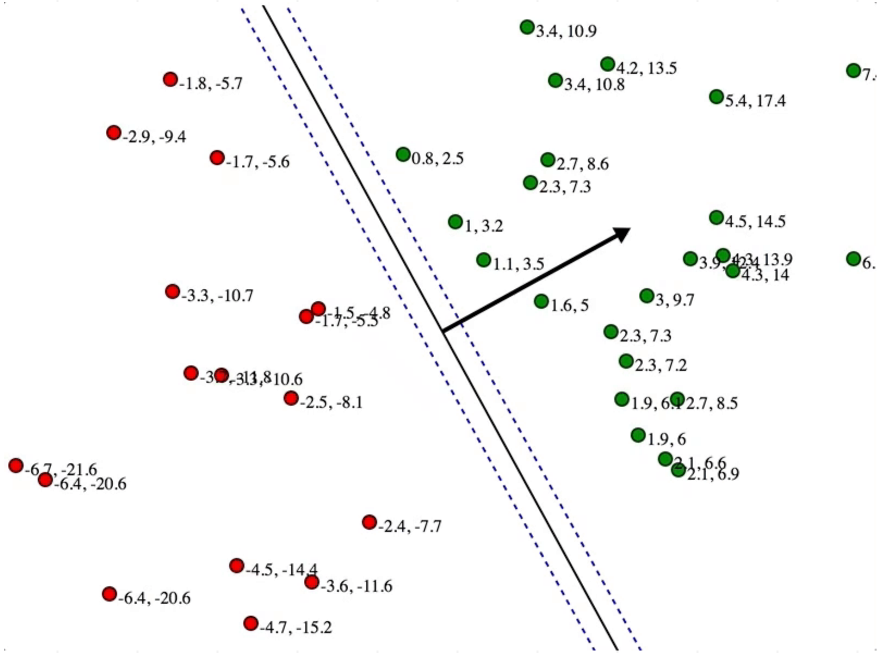
Support Vector Machine

- Here, these numbers represent distances to the line

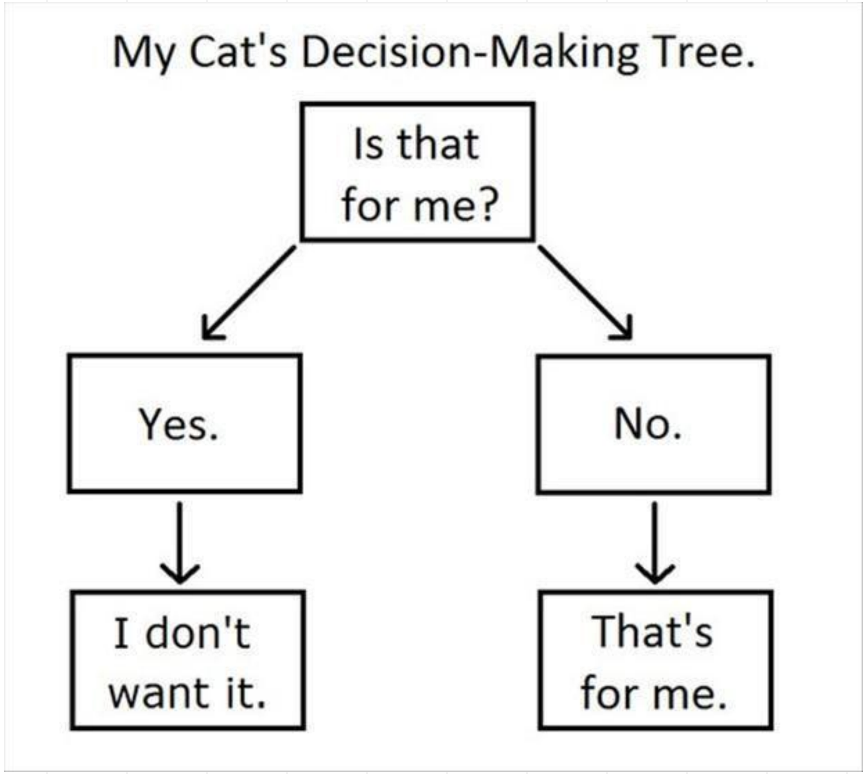
Initial State:

- The algorithm starts with an initial guess for the hyperplane.

- The algorithm iteratively adjusts the position and orientation of the hyperplane. Tweaking the parameters to improve the separation.



Decision Trees



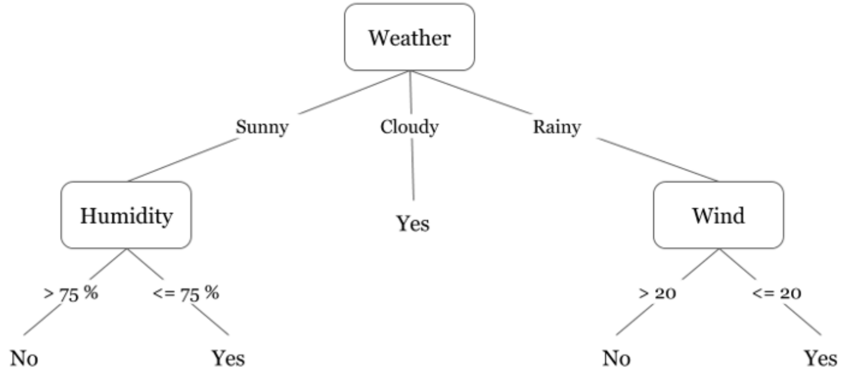
Decision Trees

Decision Tree:

- a model that makes decisions by splitting data into subsets based on feature values
- each node represents a feature
- each branch represents a decision rule
- each leaf node represents an outcome

Did I play badminton for each day of the week?

Weather	Humidity (%)	Wind Speed	Decision
Sunny	80	10	No
Sunny	60	5	Yes
Cloudy	70	15	Yes
Rainy	85	25	No
Rainy	70	10	Yes



Decision Trees

Choosing the Best Feature and threshold to Split:

- Gini impurity
- Entropy (Information Gain)

Recursive Splitting:

- Continue splitting at each node until stopping criteria are met (e.g., maximum depth, minimum samples per node)

Prediction

Did I play badminton for each day of the week?

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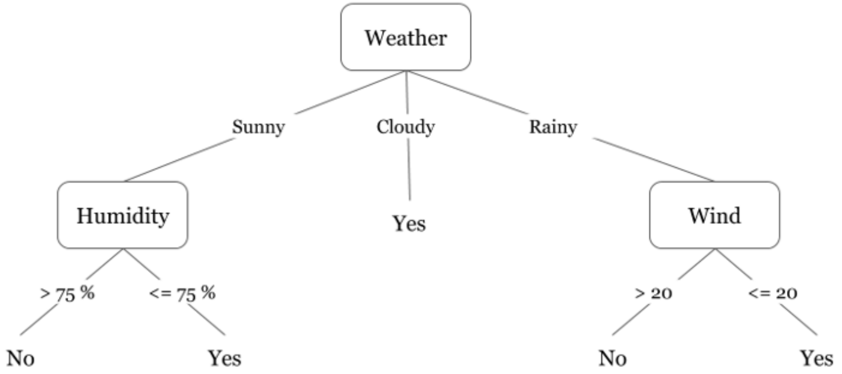

Decision Trees

Here's how we calculate Information Entropy for a dataset with C classes:

$$E = -\sum_{i}^{C} p_i \log_2 p_i$$

where p_i is the probability of randomly picking an element of class i (i.e. the proportion of the dataset made up of class i).

The Entropy of the root node:

There are 2 "No" and 3 "Yes" decisions.

$$p_{No} = \frac{2}{5}, p_{Yes} = \frac{3}{5}$$

Entropy $H(D) = -\left(\frac{2}{5}\log_2\frac{2}{5} + \frac{3}{5}\log_2\frac{3}{5}\right) \approx 0.971$

Weather	Humidity (%)	Wind Speed	Decision
Sunny	80	10	No
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Rainy	70	10	Yes

Weighted average entropy for weather:

$$H_{Weather}(D) = 0.8$$

Information Gain

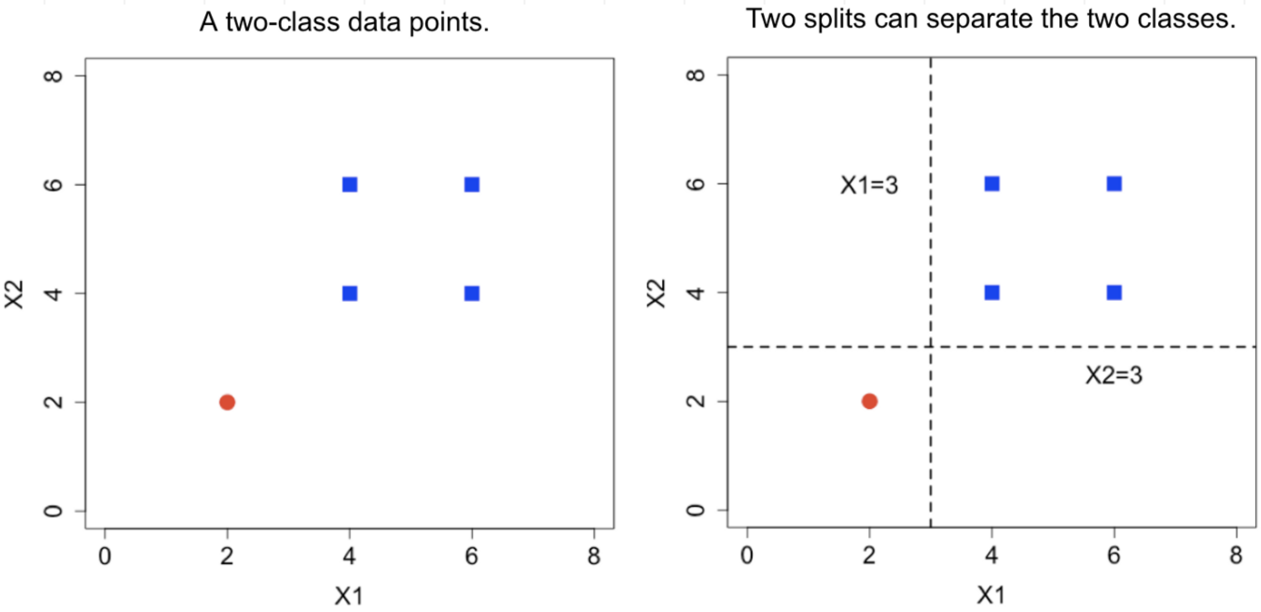
$$IG(D, Weather) = H(D) - H_{Weather}(D)$$

Decision Trees

First Split (X1 = 3):

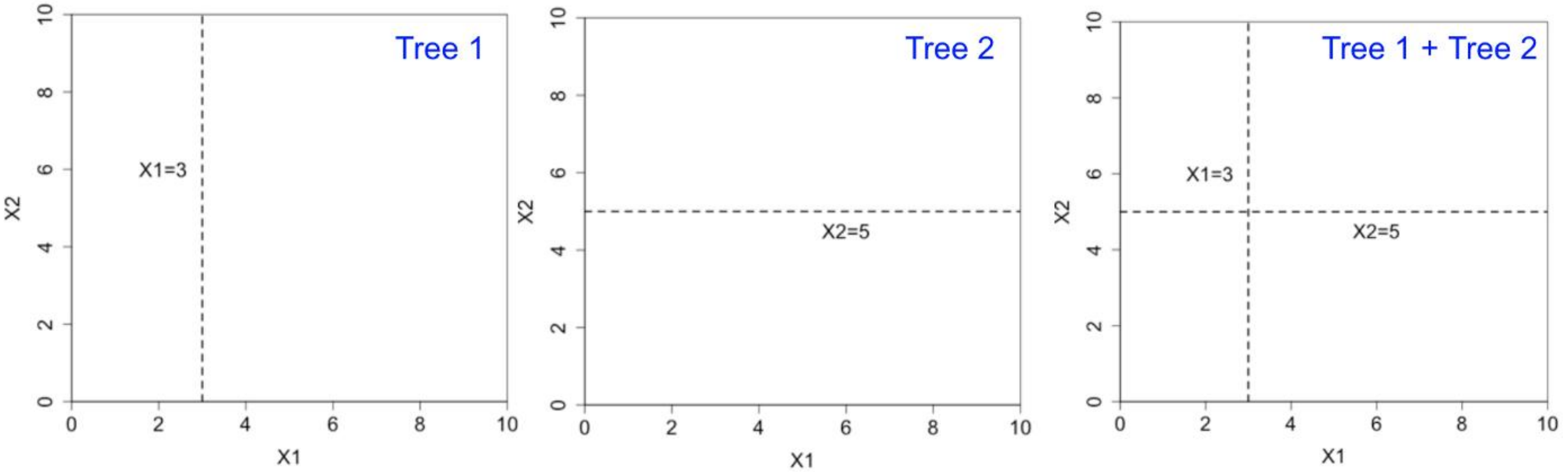
Second Split (X2 = 3)

This is another way to visualize the decision tree



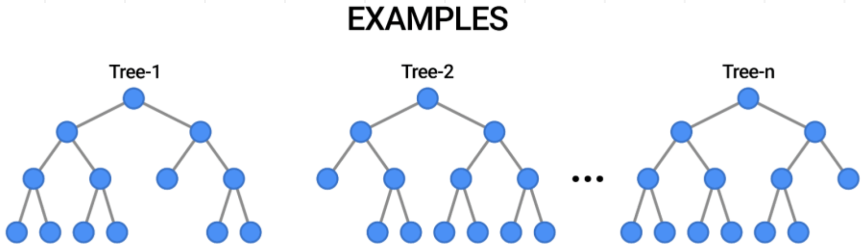
Ensemble Methods: combine the predictions of multiple models

Imagine if we had multiple (typically shallow) trees



Ensemble Methods: Random Forest

- Tree in the random forest is trained on a subset of the data and a random subset of the input features.
- So, each tree makes decisions based on a different subset of features.
- To predict a new data point, the random forest takes the majority vote of the predictions of all the individual decision trees.



Python and Machine Learning

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Python and Machine Learning

Machine Learning

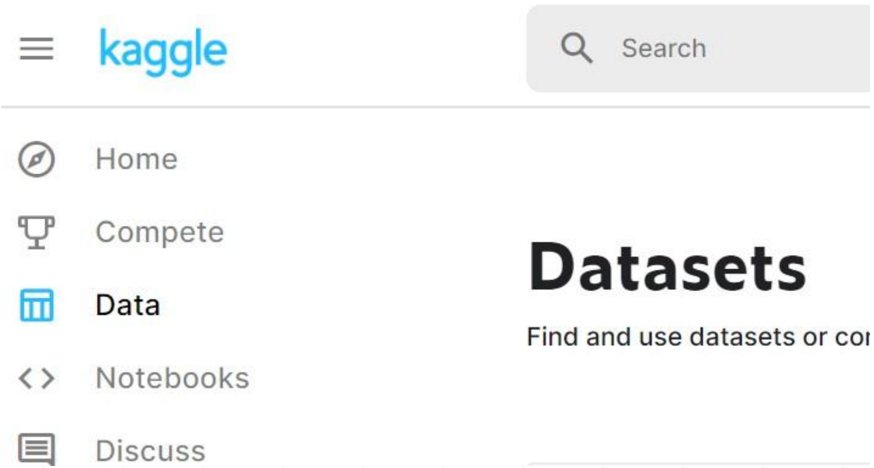
scikit learn

Deep Learning

TensorFlow

Keras

Machine Learning Datasets

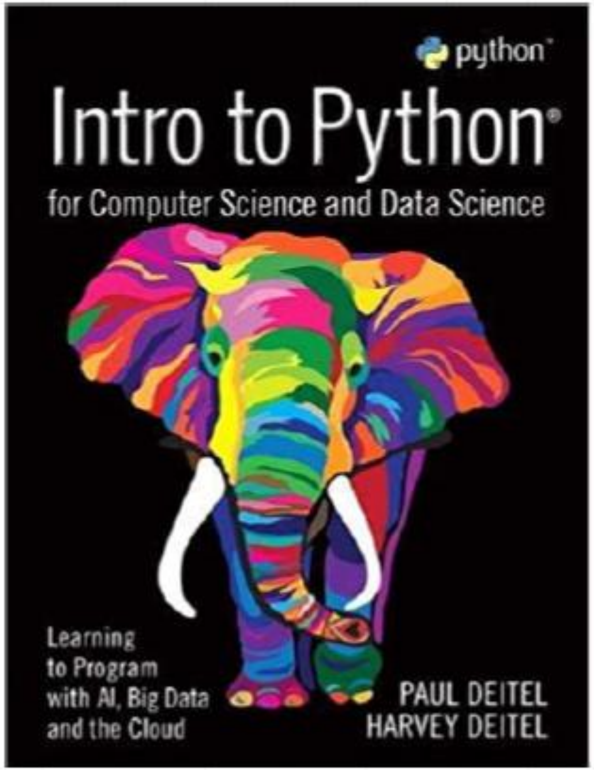


https://www.kaggle.com/datasets

https://www.openml.org/

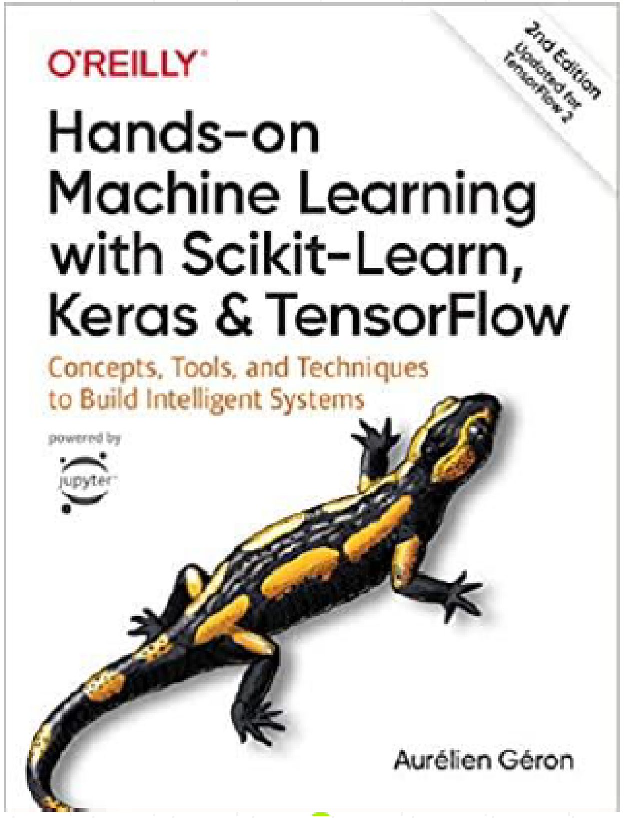
Online Courses

- Machine Learning Crash Course with TensorFlow APIs
https://developers.google.com/machine-learning/crash-course



- *Intro to Python for Computer Science and Data Science: Learning to Program with AI, Big Data and The Cloud*

  Paul J. Deitel (Author), HarveyDeitel (Author)



- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*

  Aurélien Géron (Author)

Q&A

Any questions?

You can find me at

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